HOTEL DATABASE

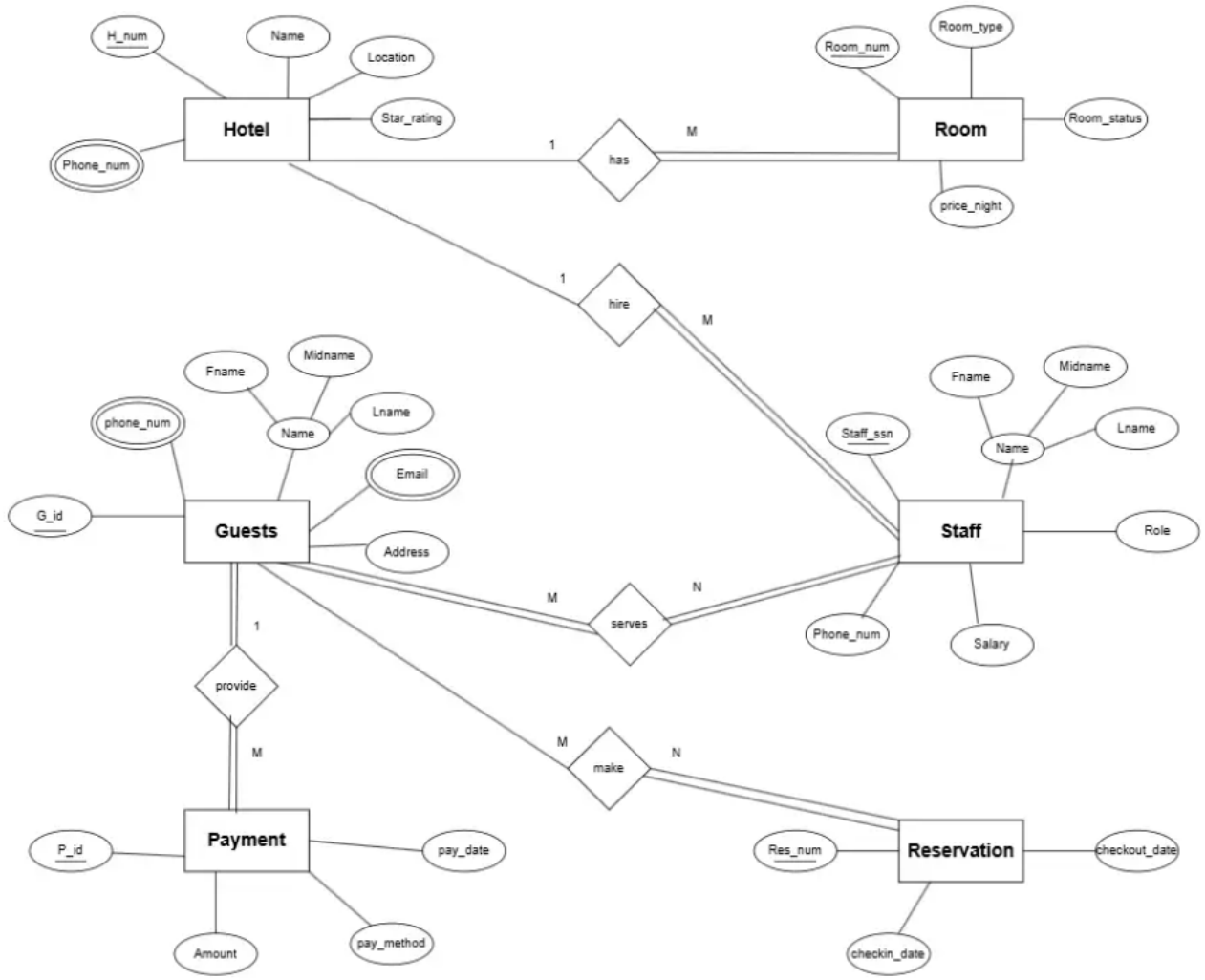
GROUP MEMBERS:

1) LAILA TAREK 231003520
2) HANA TARIQ 231000481
3) ABDALLAH NIZAR 231003010
4) HAZEM MOHAB 231000473

Database Project:

A Hotel Database System is a structured solution for managing hotel operations efficiently. It stores and organizes information about guests, rooms, reservations, staff, and services. The system ensures seamless coordination of bookings, room availability, check-ins, check-outs, and billing processes. By integrating these components, the database supports smooth day-to-day operations and enhances the overall guest experience.

ERD:



Schema:

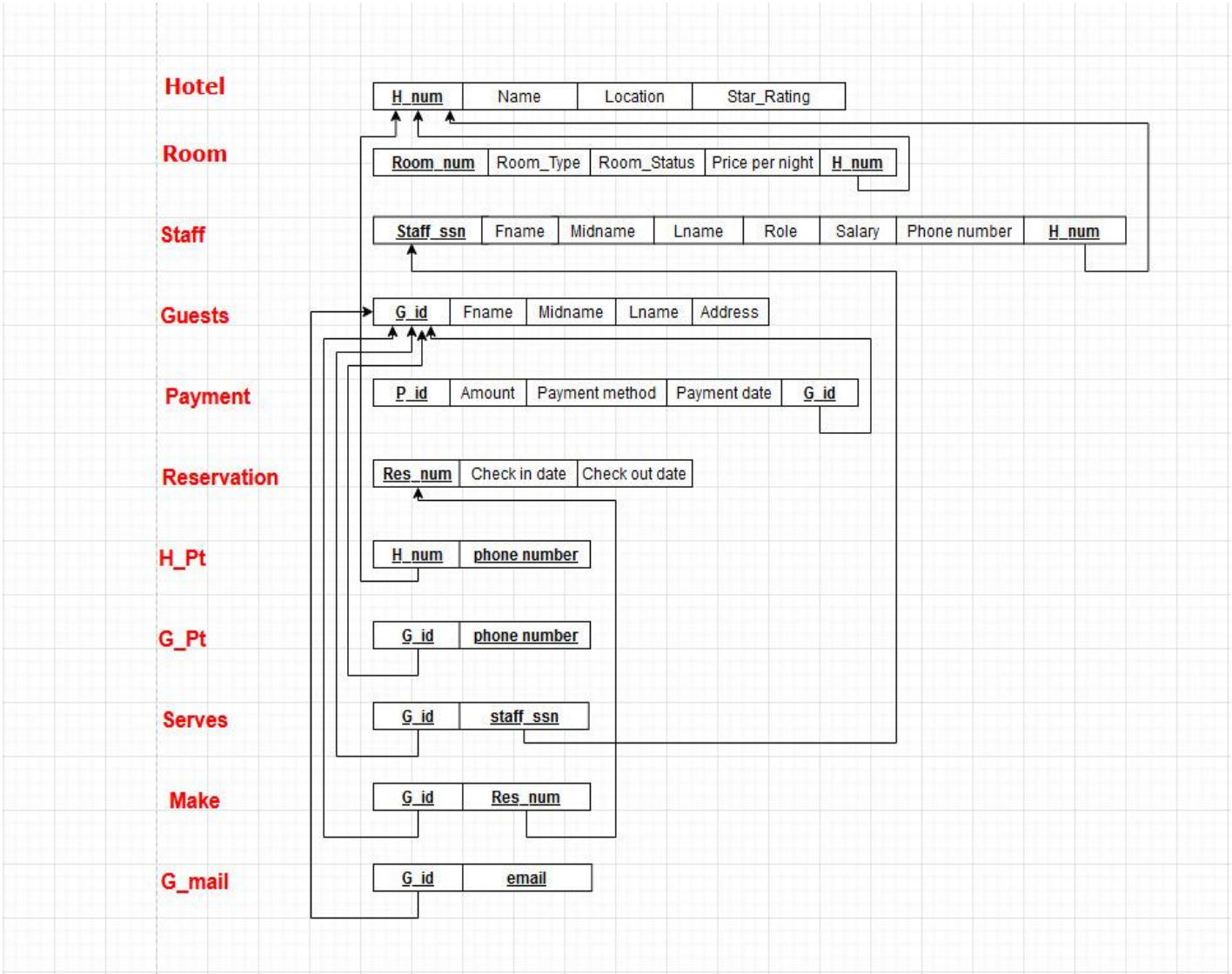

Hotel
H_num
Name
Location
Star_Rating
Room
Room_num
Room_Type
Room_Status
Price per night
H_num
Staff
Staff_ssn
Fname
Midname
Lname
Role
Salary
Phone number
H_num
Guests
G_id
Fname
Midname
Lname
Address
Payment
P_id
Amount
Payment method
Payment date
G_id
Reservation
Res_num
Check in date
Check out date
H_Pt
H_num
phone number
G_Pt
G_id
phone number
Serves
G_id
staff_ssn
Make
G_id
Res_num
G_mail
G_id
email

Table creation Queries:

SQL syntax for "Hotel" table:

```
CREATE TABLE 'Hotel' (
        H_num  INT PRIMARY KEY,
        Name VARCHAR(100) NOT NULL,
        Location VARCHAR(100) NOT NULL,
        Star_rating INT NOT NULL DEFAULT 1
);
```

SQL syntax for "Room" table:

```
CREATE TABLE 'Room'(
        Room _num  INT PRIMARY KEY,
        RoomType VARCHAR(50) NOT NULL DEFAULT 'Single',
        RoomStatus VARCHAR(50) NOT NULL DEFAULT 'Available',
        PricePerNight DECIMAL(5,2) NOT NULL DEFAULT 100.00,
        H_number INT NOT NULL,
        FOREIGN KEY (H_number) REFERENCES Hotel (H_num)
);
```

SQL syntax for "Staff" table:

```
CREATE TABLE 'Staff'(
        Staff_ssn INT PRIMARY KEY,
        Fname VARCHAR(50) NOT NULL,
        Mname VARCHAR(50),
        Lname VARCHAR(50) NOT NULL,
        Role VARCHAR(50) NOT NULL DEFAULT 'Receptionist',
        Salary DECIMAL(7,2) NOT NULL DEFAULT 10000.00,
        PhoneNumber VARCHAR(15),
        H_number INT NOT NULL,
        FOREIGN KEY (H_number) REFERENCES Hotel(H_num)
);
```

SQL syntax for “Guests” table:

```
CREATE TABLE ‘Guests’(
        G_ID INT PRIMARY KEY,
        Fname VARCHAR(50) NOT NULL,
        Mname VARCHAR(50),
        Lname VARCHAR(50) NOT NULL,
        address VARCHAR(200)
);
```

SQL syntax for “Payment” table:

```
CREATE TABLE ‘Payment’(
        P_ID INT PRIMARY KEY,
        amount DECIMAL(5,2) NOT NULL DEFAULT 100.00,
        PaymentMethod VARCHAR(50) NOT NULL DEFAULT ‘Cash’,
        PaymentDate  DATE  NOT NULL,
        Guest_id INT NOT NULL,
        FOREIGN KEY (Guest_id) REFERENCES Guests(G_ID)
);
```

SQL syntax for “H_Pt” table:

```
CREATE TABLE ‘H_Pt’(
        H_number INT ,
        Phone_number VARCHAR(50),
        PRIMARY KEY(H_number, Phone_number),
        FOREIGN KEY (H_number) REFERENCES Hotel(H_num)

);
```

SQL syntax for “G_Pt” table:

```
CREATE TABLE ‘G_Pt’(
        Guest_id INT ,
        Phone_number VARCHAR(50),
        PRIMARY KEY(Guest_id, Phone_number),
        FOREIGN KEY (Guest_id) REFERENCES Guests(G_ID)

);
```

SQL syntax for “G_email” table:

```
CREATE TABLE ‘G_email’(
        Guest_id INT ,
        email VARCHAR(50),
        PRIMARY KEY(Guest_id, email),
        FOREIGN KEY (Guest_id) REFERENCES Guests(G_ID)

);
```

SQL syntax for “Reservation” table:

```
CREATE TABLE ‘Reservation’(
        Res_num INT PRIMARY KEY ,
        CheckinDate DATE NOT NULL,
        CheckoutDate DATE NOT NULL

);
```

SQL syntax for “Make” table:

```
CREATE TABLE ‘Make’(
        Guest_id INT ,
        Reservation_num INT,
        PRIMARY KEY(Guest_id, Reservation_num),
        FOREIGN KEY(Guest_id) REFERENCES Guests(G_ID),
        FOREIGN KEY(Reservation_num) REFERENCES
Reservation(Res_num)
);
```

SQL syntax for “Serves” table:

```
CREATE TABLE ‘Serves’(
        Guest_id INT ,
        Staff_SSN INT,
        PRIMARY KEY(Guest_id, Staff_SSN),
        FOREIGN KEY(Guest_id) REFERENCES Guests(G_ID),
        FOREIGN KEY(Staff_SSN) REFERENCES Staff(Staff_ssn)
);
```

Table insertion Queries:

“Hotel” table before insertion:

H_num	Name	Location	Star_rating

SQL syntax for insertion in ’Hotel’ table:

INSERT INTO Hotel VALUES

(1, 'Four Seasons', 'Alexandria', 5),

(2, 'Hilton', 'King', 4),

(3, 'Helnan', 'Alexandria', 3),

(4, 'Tolip', 'Cairo', 5),

(5, 'Sunrise', 'Alexandria', 2);

“Hotel” table after insertion:

+ Options

←T→				H_num	Name	Location	Star_rating
Edit	Copy	Delete		1	Four Seasons	Alexandria	5
Edit	Copy	Delete		2	Hilton	King	4
Edit	Copy	Delete		3	Helnan	Alexandria	3
Edit	Copy	Delete		4	Tolip	Cairo	5
Edit	Copy	Delete		5	Sunrise	Alexandria	2

“Room” table before insertion:

Room_num	RoomType	RoomStatus	PricePerNight	H_number

Query results operations

SQL syntax for insertion in ’Room’ table:

INSERT INTO Room VALUES

(101, 'Single', 'Available', 120.00, 1),

(102, 'Double', 'Occupied', 200.00, 2),

(201, 'Suite', 'Available', 300.00, 3),

(202, 'Single', 'Maintenance', 100.00, 4),

(301, 'Double', 'Available', 220.00, 5);

“Room” table after insertion:

	Room_num	RoomType	RoomStatus	PricePerNight	H_number
Edit Copy Delete	101	Single	Available	120.00	1
Edit Copy Delete	102	Double	Occupied	200.00	2
Edit Copy Delete	201	Suite	Available	300.00	3
Edit Copy Delete	202	Single	Maintenance	100.00	4
Edit Copy Delete	301	Double	Available	220.00	5

“Staff” table before insertion:

Staff_ssn	Fname	Mname	Lname	Role	Salary	PhoneNumber	H_number

Query results operations

SQL syntax for insertion in ’Staff’ table:

INSERT INTO Staff VALUES

(101001, 'Hazem', 'M', 'Ragab', 'Manager', 50000.00, '1234567890', 1),

(101002, 'Laila', NULL, 'Tarek', 'Receptionist', 25000.00, '0987654321', 4),

(101003, 'Hana', NULL, 'Tarek', 'Chef', 35000.00, '1122334455', 2),

(101004, 'Abdallah', NULL, 'Nizar', 'Housekeeper', 20000.00, '6677889900', 3),

(101005, 'Eyad', NULL, 'Ehab', 'Security', 30000.00, '3344556677', 1);

“Staff” table after insertion:

	Staff_ssn	Fname	Mname	Lname	Role	Salary	PhoneNumber	H_number
Edit Copy Delete	101001	Hazem	M	Ragab	Manager	50000.00	1234567890	1
Edit Copy Delete	101002	Laila	NULL	Tarek	Receptionist	25000.00	0987654321	4
Edit Copy Delete	101003	Hana	NULL	Tarek	Chef	35000.00	1122334455	2
Edit Copy Delete	101004	Abdallah	NULL	Nizar	Housekeeper	20000.00	6677889900	3
Edit Copy Delete	101005	Eyad	NULL	Ehab	Security	30000.00	3344556677	1

“Guests” table before insertion:

G_ID	Fname	Mname	Lname	address

Query results operations

SQL syntax for insertion in ’Guests’ table:

INSERT INTO Guests VALUES

(1, 'Mahmoud', 'A', 'Karim', '123 Main St, NY'),

(2, 'Bob', NULL, 'Miller', '456 Elm St, CA'),

(3, 'Charlie', 'K', 'Evans', '789 Oak St, CO'),

(4, 'Daisy', 'L', 'Walker', '101 Pine St, IL'),

(5, 'Evan', 'M', 'White', '202 Birch St, MI');

“Guests” table after insertion:

+ Options

	G_ID	Fname	Mname	Lname	address
Edit Copy Delete	1	Mahmoud	A	Karim	123 Main St, NY
Edit Copy Delete	2	Bob	NULL	Miller	456 Elm St, CA
Edit Copy Delete	3	Charlie	K	Evans	789 Oak St, CO
Edit Copy Delete	4	Daisy	L	Walker	101 Pine St, IL
Edit Copy Delete	5	Evan	M	White	202 Birch St, MI

Check all With selected: Edit Copy Delete Export

“Payment” table before insertion:

P_ID	amount	PaymentMethod	PaymentDate	Guest_id

Query results operations

SQL syntax for insertion in ’Payment’ table:

INSERT INTO Payment VALUES

(1, 150.00, 'Credit Card', '2024-12-01', 1),

(2, 200.00, 'Cash', '2024-12-02', 2),

(3, 250.00, 'Debit Card', '2024-12-03', 3),

(4, 180.00, 'Credit Card', '2024-12-04', 4),

(5, 300.00, 'Cash', '2024-12-05', 5);

“Payment” table after insertion:

+ Options

		P_ID	amount	PaymentMethod	PaymentDate	Guest_id
Edit Copy Delete		1	150.00	Credit Card	2024-12-01	1
Edit Copy Delete		2	200.00	Cash	2024-12-02	2
Edit Copy Delete		3	250.00	Debit Card	2024-12-03	3
Edit Copy Delete		4	180.00	Credit Card	2024-12-04	4
Edit Copy Delete		5	300.00	Cash	2024-12-05	5

Check all With selected: Edit Copy Delete Export

“H_Pt” table before insertion:

H_number	Phone_number

Query results operations

Create view

SQL syntax for insertion in ’H_Pt’ table:

INSERT INTO H_Pt VALUES

(1, '010-383-6754'),

(1, '012-474-3845'),

(2, '03-483-3333'),

(2, '03-581-8000'),

(3, '222-222-2222');

“H_Pt” table after insertion:

				H_number	Phone_number
Edit	Copy	Delete		1	010-383-6754
Edit	Copy	Delete		1	012-474-3845
Edit	Copy	Delete		2	03-483-3333
Edit	Copy	Delete		2	03-581-8000
Edit	Copy	Delete		3	222-222-2222

“G_Pt” table before insertion:

Guest_id	Phone_number

SQL syntax for insertion in ’G_Pt’ table:

INSERT INTO G_Pt VALUES

(1, '012-7646-766'),

(1, '011-909-7612'),

(3, '966-233-0998'),

(3, '010-000-1942'),

(5, '999-456-4327');

“G_Pt” table after insertion:

+ Options

				Guest_id	Phone_number
Edit	Copy	Delete		1	011-909-7612
Edit	Copy	Delete		1	012-7646-766
Edit	Copy	Delete		3	010-000-1942
Edit	Copy	Delete		3	966-233-0998
Edit	Copy	Delete		5	999-456-4327

“G_email” table before insertion:

```
SELECT * FROM `G_email`
```

Guest_id	email

SQL syntax for insertion in ’G_email’ table:

INSERT INTO G_email VALUES

(1, 'mahmoud67@gmail.com'),

(1, 'mahmoud@hotmail.com'),

(3, 'charlie@gmail.com'),

(3, 'charlie@outlook.com'),

(5, 'evan@gmail.com');

“G_email” table after insertion:

+ Options

			Guest_id	email
Edit	Copy	Delete	1	mahmoud67@gmail.com
Edit	Copy	Delete	2	bob@hotmail.com
Edit	Copy	Delete	3	charlie@gmail.com
Edit	Copy	Delete	4	daisy@outlook.com
Edit	Copy	Delete	5	evan@gmail.com

“Reservation” table before insertion:

Res_num	CheckinDate	CheckoutDate

SQL syntax for insertion in ’Reservation’ table:

INSERT INTO Reservation VALUES

(1, '2024-12-10', '2024-12-15'),

(2, '2024-12-11', '2024-12-16'),

(3, '2024-12-12', '2024-12-17'),

(4, '2024-12-13', '2024-12-18'),

(5, '2024-12-14', '2024-12-19');

“Reservation” table after insertion:

Options

			Res_num	CheckinDate	CheckoutDate
Edit	Copy	Delete	1	2024-12-10	2024-12-15
Edit	Copy	Delete	2	2024-12-11	2024-12-16
Edit	Copy	Delete	3	2024-12-12	2024-12-17
Edit	Copy	Delete	4	2024-12-13	2024-12-18
Edit	Copy	Delete	5	2024-12-14	2024-12-19

Check all *With selected:* Edit Copy Delete Export

“Make” table before insertion:

Guest_id	Reservation_num

SQL syntax for insertion in ’Make’ table:

INSERT INTO Make VALUES

(1, 1),

(2, 2),

(3, 3),

(4, 4),

(5, 5);

“Make” table after insertion:

Show all | Number of rows: 25 Filter rows: Search t

Options

			Guest_id	Reservation_num
Edit	Copy	Delete	1	1
Edit	Copy	Delete	2	2
Edit	Copy	Delete	3	3
Edit	Copy	Delete	4	4
Edit	Copy	Delete	5	5

“Serves” table before insertion:

Guest_id	Staff_SSN

SQL syntax for insertion in ’Serves’ table:

INSERT INTO Serves VALUES

(1, 101001),

(2, 101002),

(3, 101003),

(4, 101004),

(5, 101005);

“Serves” table after insertion:

+ Options

←T→			Guest_id	Staff_SSN
Edit	Copy	Delete	1	101001
Edit	Copy	Delete	2	101002
Edit	Copy	Delete	3	101003
Edit	Copy	Delete	4	101004
Edit	Copy	Delete	5	101005

Bonus Task:

1st Subquery:

```
SELECT Fname, Lname, address
FROM Guests
WHERE Fname LIKE '%a%';
```

Result:

+ Options

			Fname	Lname	address
Edit	Copy	Delete	Mahmoud	Karim	123 Main St, NY
Edit	Copy	Delete	Charlie	Evans	789 Oak St, CO
Edit	Copy	Delete	Daisy	Walker	101 Pine St, IL
Edit	Copy	Delete	Evan	White	202 Birch St, MI

2nd Subquery:

```
SELECT AVG(Salary)
FROM Staff;
```

Result:

+ Options

AVG(Salary)
32000.000000

3rd Subquery:

```
SELECT COUNT(G_ID)
FROM Guests;
```

Result:

+ Options

COUNT(G_ID)
5